

501 → Amrad Sir 8nA → 18 to 22,

Q12. Proof that If a preference relation satisfies monotonicity and continuity then there exists a utility function $u(\cdot)$ representing such preference relation.

1) Ans: Let us take a bundle $x \neq 0$

By monotonicity, $x \succ 0$, where $0 = (0, 0, \dots, 0)$. Indeed, if bundle $x \neq 0$, then it contains positive amounts of at least one good, and, by monotonicity it is preferred to bundle 0.

Let $m \equiv \max \{x_1, x_2, \dots, x_n\}$ be the number of units of the most abundant good in bundle x . Let us define bundle M as the bundle where all components coincide with the highest component of bundle x . That is, $M = (m, m, \dots, m)$

Bundle M hence increases the amounts of all goods in bundle except for the most abundant commodity, good 1 in the followin fig. And hence, by monotonicity we have that consumer prefers M to x , $M \succ x$.

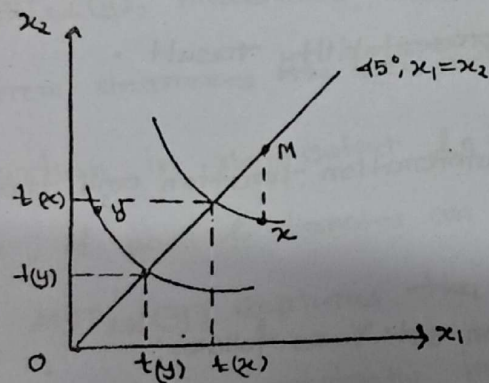


Fig: Existence of utility function.

Bundles 0 and M are both on the main diagonal, since each of them contains the same amount of good x_1 and x_2 . By continuity and monotonicity, there exists a bundle that is indifferent to x and lies on the main diagonal. By monotonicity, this bundle is unique.

we denote this bundle as $(t(x), t(x), \dots, t(x))$. For completeness, let us define $u(x) = t(x)$, which is a real number.

Applying the same steps for another bundle $y \in X$, we obtain a similar bundle on the same diagonal, $(t(y), t(y), \dots, t(y))$, which lies on the same indifference curve as bundle y . For completeness, let $u(y) = t(y)$, which is also a real number. We know that,

$$x \sim (t(x), t(x), \dots, t(x)), y \sim (t(y), t(y), \dots, t(y)) \text{ and, } x \succeq y$$

Hence, by transitivity of preference relation $x \succeq y$, if and only if

$$x \sim (t(x), t(x), \dots, t(x)) \succeq (t(y), t(y), \dots, t(y)) \sim y$$

and by monotonicity, this is true as long as the real number $t(x)$ exceeds $t(y)$, meaning $t(x) \geq t(y)$,

$$x \succeq y \Leftrightarrow t(x) \geq t(y) \Leftrightarrow u(x) \geq u(y)$$

as required for our representability result. 2

Q.18 Define production vector/transformation function and transformation frontier.

Ans: If we consider a production set, Y as follows:

$$Y = \{y \in \mathbb{R}^L : F(y) \leq 0\}$$

here, $F(y)$ is the transformation function. This function can be intuitively understood as a production function as following fig illustrates:

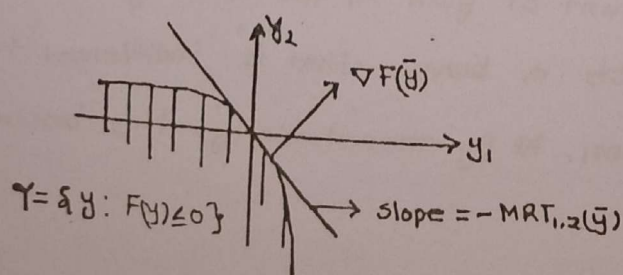


Fig: Production set and transformation

Here, the boundary of the production function indicates production plans for which $F(y) = 0$. This boundary is the transformation frontier. Points strictly below the transformation frontier indicate (strictly) feasible production plans since $F(y) < 0$. Therefore we can totally differentiate $F(y)$ as follows:

$$dF(y) = \frac{\partial F(y)}{\partial y_k} dy_k + \frac{\partial F(y)}{\partial y_l} dy_l$$

Now, we evaluate the expression above at any point $y = \bar{y}$ on the transformation frontier where $dF(\bar{y}) = 0$ holds. Solving for dy_l / dy_k , we

obtain
$$\frac{dy_l}{dy_k} = - \frac{\partial F(\bar{y}) / \partial y_k}{\partial F(\bar{y}) / \partial y_l}$$

where the term on R.H.S is $\frac{\partial F(\bar{y}) / \partial y_k}{\partial F(\bar{y}) / \partial y_l} = MRT_{lk}(\bar{y})$

The marginal rate of transformation between good l and k , evaluated at $\bar{y}$, $MRT_{lk}(\bar{y})$, measures how much the (net) output k can increase if the firm decreases the (net) output of good l by one unit, where the reduction in net output l implies a larger input usage. For instance, if good l denotes an input while k represents an output, $MRT_{lk}(\bar{y})$ captures the increase in output y_k that the firm obtains from marginally increasing the amount of input y_l , it employs.

Q12. Define production set. Explain the properties of production set.

Ans: Let us define a production plan $y = (y_1, y_2, \dots, y_L)$ as a vector with L components. If a particular component of the vector is

positive, it denotes that the firm produces positive amounts of good; for instance $y_1 > 0$. If instead a component of the y vector is negative, it denotes that the firm uses good 2 as an input in the production process. A production plan that is technologically feasible is represented as a point of production set Y , as follows:

$$Y = \{y \in \mathbb{R}^L : F(y) \leq 0\} ; \text{ where } F(y) \text{ is the transformation function}$$

Different properties of production sets are given below:

1. Y is nonempty: That is, we have inputs and/or outputs. ~~For~~ If a firm, despite using large quantities of inputs produces no outputs our production set will still be nonempty. This case is illustrated in following fig where the production set Y is nonempty, despite producing no units of outputs.

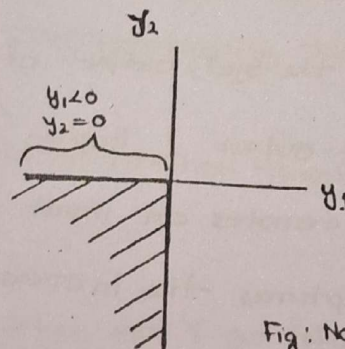


Fig: Non-empty production set Y

2. Y is closed: Hence the production set Y includes its boundary points.
3. No free lunch: This property states that firm must use inputs in order to produce output. The top panel of the following fig represents a production set that satisfies no free lunch property, since the firm is using amounts of input y_1 in order to produce positive

amounts of output y_2 . In contrast, the bottom panels of the fig illustrate production plans that violate the "no-free lunch" property.

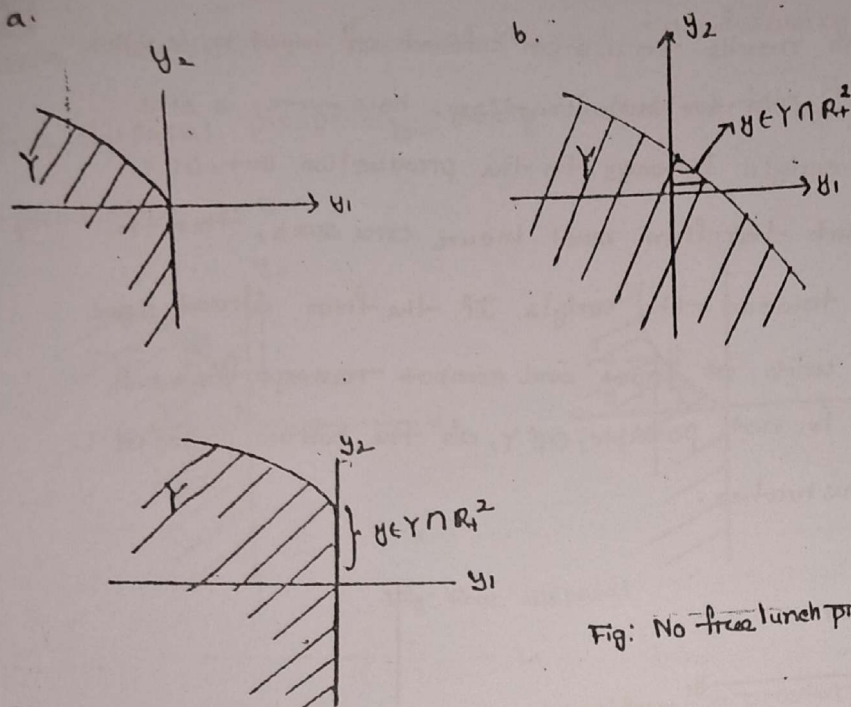


Fig: No free lunch property.

4. Possibility of inaction: This property states that firm can choose to be inactive, using no inputs and obtaining no output as a consequence. At first In other words, the input-output vector 0 is part of the production set. At first glance, a firm's possibility of inaction may seem applicable to many production processes, such as the production set depleted in the following fig:

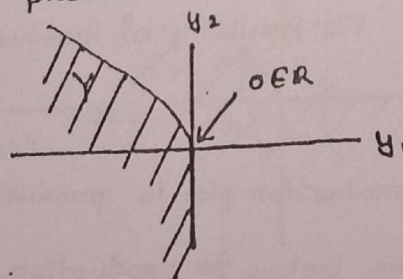


Fig: Possibility of inaction-I.

However, when firms face sunk costs, this property might be violated, meaning $0 \notin Y$. In particular, if the firm needs to use

experiences a setup cost (fixed) as the top panel of the following fig depicts, the firm needs to use an amount of input y_1 without obtaining any output y_2 in return. Inaction, however, is still possible since the origin belongs to the production set, $O \in Y$. If the fixed costs that the firm must incur are sunk, then the firm cannot move toward the origin. If the firm already signed the purchase of y_2 units of input and cannot renegotiate from such a contract, inaction is not possible, $O \notin Y$, as the bottom panel of the following fig illustrates.

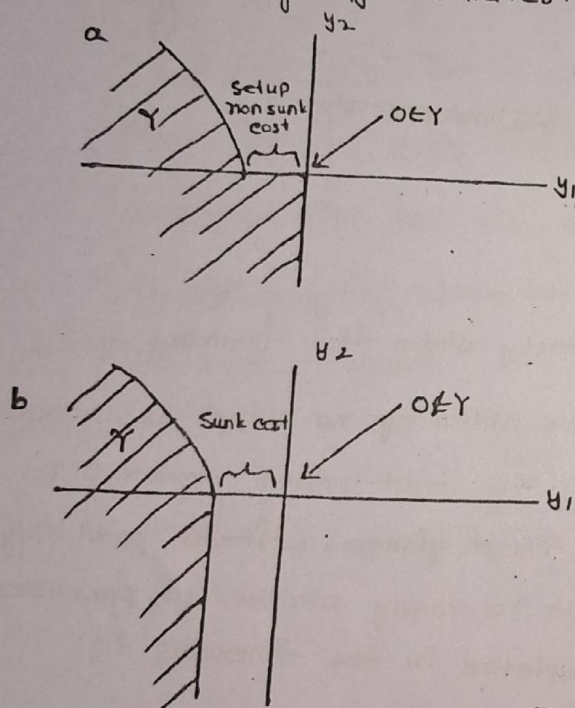


Fig: Possibility of Inaction-II

5. Free disposal: If y is a production plan in production set Y , $y \in Y$, and $y' \leq y$, then y' must also belong to production set Y . Intuitively, production plan y' is less efficient than production plan y . The production plan y' must also belong to the firm's production set. Hence the producer can use more inputs without

the need to reduce its output, as he can dispose of the additional inputs he does not need at no cost. The following fig illustrates the free disposal property being satisfied for two very different production sets.

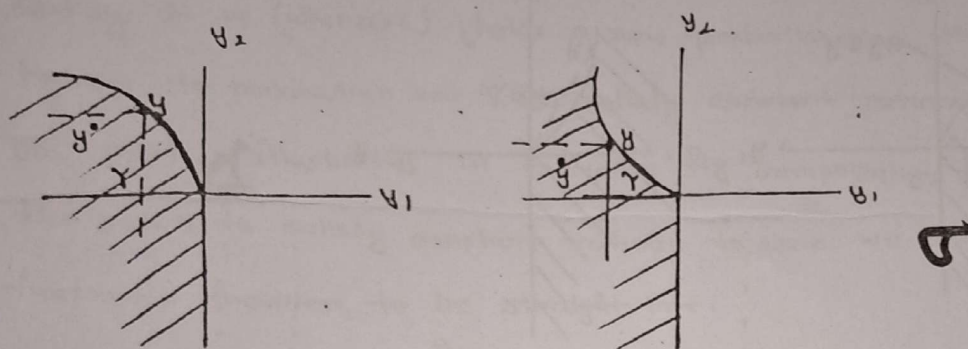
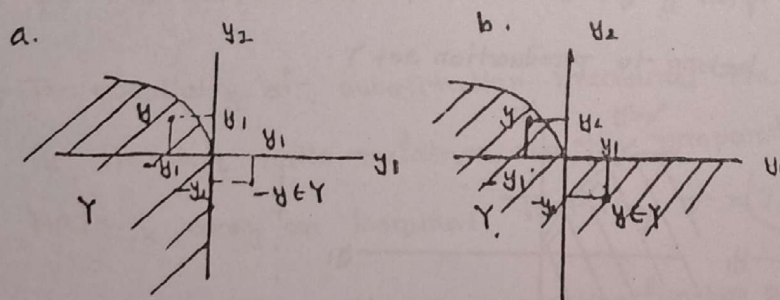


Fig: Free disposal

6. Irreversibility: Let us consider a production plan $y \in Y$, where y does not coincide with the origin, that is, $y \neq 0$. Then, production plan $-y$ cannot belong to Y . The following fig illustrates the no irreversibility property. This property highlights that there is no way back for this firm, in sense of transforming all its output into the exact amount of inputs that generated such output.



7. Non-increasing returns to scale: Let us take a production plan $y \in Y$. Then any scaling down of this production plan $y, \alpha y$, where $\alpha \in [0, 1]$ is also a part of production set. The following fig illustrates a production

set satisfying nonincreasing returns to scale in the left panel and a production set that violates non-increasing returns to scale on the right panel.

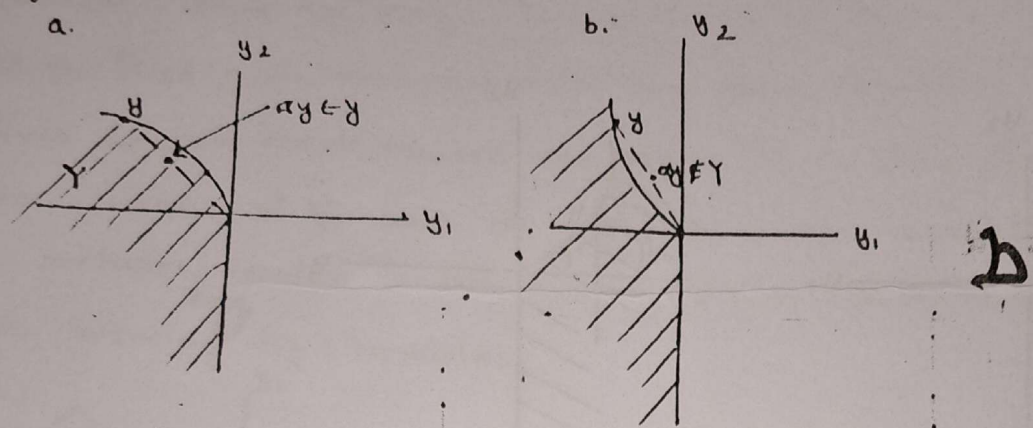


Fig: Non-increasing returns to scale

8. Non-decreasing returns to scale: Let us take a production plan $y \in Y$. Then any scaling up of this production plan αy , where $\alpha \geq 1$, is also part of production set Y . The following fig (left hand panel) depicts a production set satisfying non-decreasing returns to scale, since scaling up any production plan y yields a new production plan αy that also belongs to Y . In contrast the right hand panel shows a production set that violates non-decreasing returns to scale:

scaling up production plan y yields a new production plan αy that doesn't necessarily belong to production set Y .

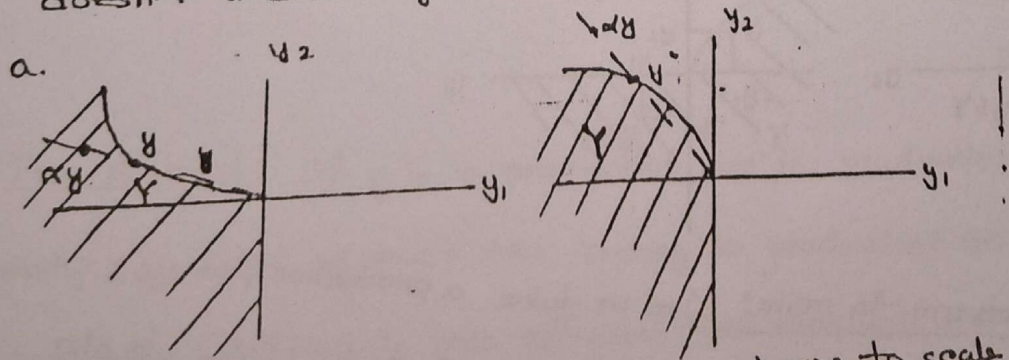


Fig: Non-decreasing returns to scale.

9. Constant returns to scale: Let us take a production plan $y \in Y$. Then production plan αy also belongs to production set Y , for any $\alpha \geq 0$. Hence both scaling down an original production plan y (when $\alpha \in [0, 1]$) and scaling it up (when $\alpha > 1$) yields a new production plan that still belongs to production set Y . to satisfy constant returns to scale

This point is illustrated in following fig. Graphically, for a production set Y to satisfy constant returns to scale, we need the transformation frontier to be straight line.

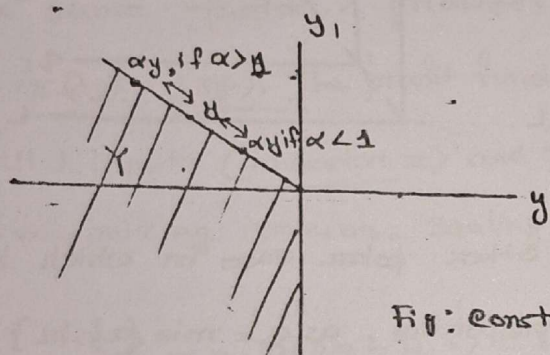


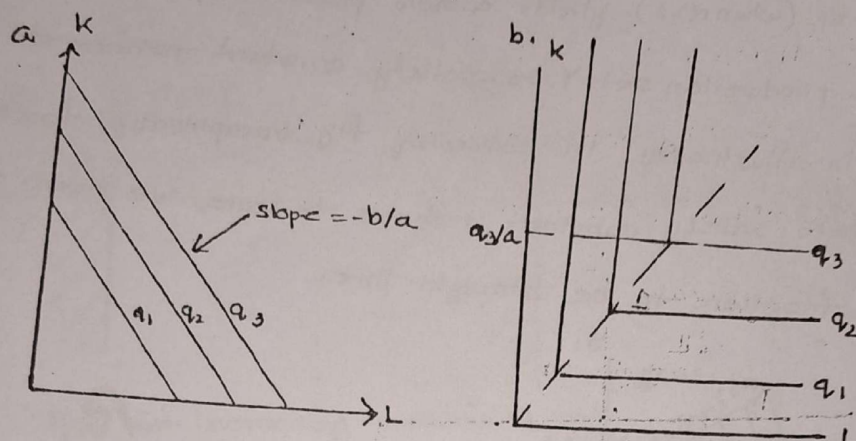
Fig: Constant returns to scale.

Q.20 Obtain the elasticity of substitution (E_{σ}). Find the Elasticities of substitution for the production function $q = f(k, L) = ak + bL$. What will be the EOS of the production function in fixed proportion?

Ans: The elasticity of substitution measures the proportionate change in the k/L ratio relative to the proportionate change in the $MRTS_{L,k}$ along an isoquant. That is, $\sigma = \left(\% \Delta \frac{k}{L} \right) / \left(\% \Delta MRTS_{L,k} \right)$

Let us consider a linear production function $q = f(k, L) = ak + bL$. Solving for k in production function $q = ak + bL$ yields an isoquant $k = q/a - (b/a)L$, with slope $-b/a$, which is constant in labor.

Panel a of the following fig illustrates a linear production function in which labor and capital are perfect substitutes in production process. It implies that $MRTS_{L,k}$ is constant in k/L and hence the elasticity of substitution yields an infinitely large number $\sigma = \infty$.



Let us next examine the other polar case in which both inputs must be used in fixed proportion, as $q = \min\{ak, bL\}$, where $a, b > 0$. In this case the $MRTS_{L,k}$ changes dramatically from infinite (for labor amounts below the kink in the isoquants as depicted in panel b) to zero. This implies the change in $MRTS_{L,k}$ is infinite, yielding as a consequence a zero elasticity of substitution, namely $\sigma = 0$.

Q. 21. Define profit function. If the production set Y is closed and have fixed disposal, then what ^{properties} responsibilities do the profit function satisfy?

Ans: If we consider a profit maximization problem, $\max p \cdot y$ subject to $y \in Y$, the value function resulting from this profit maximization problem, $\pi(p)$, do

ted as the "profit function" of the firm, describes for every price p the highest amount of profits, chosen by the profit-maximizing production plan $y(p)$. More formally, we can define profit function as

$$\pi(p) = p \cdot y(p); \quad p \cdot y(p) = p \cdot y(p)$$

that is the firm's profit evaluated at $y = y(p)$.

Now, let us describe some properties of the profit function $\pi(p)$.

Let us assume that the production set Y is closed and satisfies the free disposal property:

1. Homogeneity: The profit function $\pi(p)$ is homogeneous of degree one in prices. That is increasing the prices of all inputs and outputs by the same factor λ produces a proportional increase in the firm's profits, $\pi(\lambda p) = \lambda \pi(p)$. The profit function can be expressed as $\pi(p) = p \cdot q - w \cdot z$, where all inputs (in vector z) and the output q are evaluated at their profit-maximizing amounts. Scaling all prices by common factor λ , we obtain

$$\lambda p \cdot q - \lambda w \cdot z = \lambda (p \cdot q - w \cdot z) = \lambda \cdot \pi(p)$$

which shows that the firm's profit increases in the same proportion (λ) as (input and output prices), thus implying that homogeneity of degree one.

2. Convex in price: The profit function $\pi(p)$ is convex. The following fig describes this property. For a price vector p , the profit-maximizing production plan is $y(p)$, yielding an associated profit function $\pi(p)$. Similarly, for a price vector p' , the profit maximizing production plan is $y(p')$, yielding an associated profit function $\pi(p')$.

3. Alternative representation: If a production set Y is convex, then

$$Y = \{y \in \mathbb{R}^L : py \leq \pi(p) \text{ for all } p \gg 0\} \quad Y = \{y \in \mathbb{R}^L : py \leq \pi(p) \text{ for all } p \gg 0\}$$

Intuitively, the production set Y can be alternatively represented by this "dual" set. It specifies that, for any given price vector p , all production vector y generate less profit py than the profit maximizing production plan $y(p)$, which entails a profit function $\pi(p) = py(p)$. A graphical representation of this property is shown in following fig:

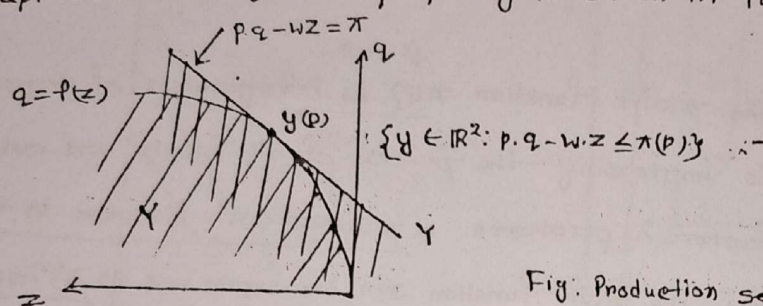


Fig. Production set Y is convex.

As the fig shows, in a convex production set Y , the supply corresponding $y(p)$ maximizes profits and the associated isoprofit line is $\pi = p q - w z$. Since all production plans (z, q) below this isoprofit line is ~~isoprofit~~ yield a lower profit level for the firm $p q - w z \leq \pi(p)$. Alternatively, the isoprofit line $\pi = p q - w z$ can be represented by

$$q = \frac{\pi}{p} - \frac{w}{p} z$$

If the price vector w/p is constant, meaning input (output) prices are unaffected by different level of input usage, then the slope of the isoprofit line is constant in z . Therefore the set of (z, q) pairs below profit function $\pi(p)$ define a convex set.

Q.22 Consider a consumer with the utility function $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$

i) Derive the warrasian and compensated demand function.

ii) Consider $\{p^0, m^0\} = \{4, 4, 300\}$ a) compute the indirect utilities and
 $\{p^1, m^0\} = \{2, 4, 300\}$ cost

iii) calculate EV, CV (S es) and examine their relationship most common on the

i) Given, Cobb-Douglas utility function, $U(x_1, x_2) = x_1^{1/3} \cdot x_2^{1/2}$

Budget constraint, $m = P_1 x_1 + P_2 x_2$

Waltzian / Marshallian demand function $\rightarrow$ Maximize utility

$$x_1(P, w) = \frac{\alpha w}{(\alpha + \beta) P_1} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{2}} \cdot \frac{m}{P_1} = \frac{\frac{1}{3}}{\frac{5}{6}} \cdot \frac{m}{P_1} = \frac{2}{5} \cdot \frac{m}{P_1}$$

$$x_2(P, w) = \frac{\beta w}{(\alpha + \beta) P_2} = \frac{\frac{1}{2}}{\frac{1}{3} + \frac{1}{2}} \cdot \frac{m}{P_2} = \frac{\frac{1}{2}}{\frac{5}{6}} \cdot \frac{m}{P_2} = \frac{3}{5} \cdot \frac{m}{P_2}$$

Compensated / Hicksian demand function $\rightarrow$ Minimize expenditure

$$h_1(P, u) = \left(\frac{\alpha}{\beta} \cdot \frac{P_2}{P_1} \right)^{\frac{\beta}{\alpha + \beta}} \cdot u^{\frac{1}{\alpha + \beta}} = \left(\frac{\frac{1}{3}}{\frac{1}{2}} \cdot \frac{P_2}{P_1} \right)^{\frac{1}{5}} \cdot u^{\frac{1}{5}} = \left(\frac{P_2}{2P_1} \right)^{\frac{1}{5}} \cdot u^{\frac{1}{5}}$$

$$h_2(P, u) = \left(\frac{\beta}{\alpha} \cdot \frac{P_1}{P_2} \right)^{\frac{\alpha}{\alpha + \beta}} \cdot u^{\frac{1}{\alpha + \beta}} = \left(\frac{\frac{1}{2}}{\frac{1}{3}} \cdot \frac{P_1}{P_2} \right)^{\frac{1}{5}} \cdot u^{\frac{1}{5}} = \left(\frac{3P_1}{P_2} \right)^{\frac{1}{5}} \cdot u^{\frac{1}{5}}$$

ii) Indirect Utility:

$$\text{Indirect utility } U = v(P, w) = \left(\frac{\alpha}{\alpha + \beta} \cdot \frac{w}{P_1} \right)^{\alpha} \left(\frac{\beta}{\alpha + \beta} \cdot \frac{w}{P_2} \right)^{\beta} = \left(\frac{\alpha}{P_1} \right)^{\alpha} \left(\frac{\beta}{P_2} \right)^{\beta} \left(\frac{w}{\alpha + \beta} \right)^{\alpha + \beta}$$

$$\text{Indirect utility at initial prices } U^0 = \left(\frac{1/3}{1} \right)^{1/3} \left(\frac{1/2}{1} \right)^{1/2} \left(\frac{300}{1/3 + 1/2} \right)^{5/6} = 15.81$$

$$\text{Indirect } \gg \gg \text{ final } \gg U^1 = \left(\frac{1/3}{2} \right)^{1/3} \left(\frac{1/2}{1} \right)^{1/2} \left(\frac{300}{1/3 + 1/2} \right)^{5/6} = 18.80$$

Costs / Expenditure:

$$\text{Costs } e(P, u) = P_1^{\frac{\alpha}{\alpha + \beta}} P_2^{\frac{\beta}{\alpha + \beta}} u^{\frac{1}{\alpha + \beta}} \left[\left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha + \beta}} + \left(\frac{\beta}{\alpha} \right)^{\frac{\alpha}{\alpha + \beta}} \right]$$

Costs under initial prices, P^0 and maximal utility level $U^0 \equiv v(P^0, w)$

$$e(P^0, U^0) = 1^{\frac{1/3}{1/3 + 1/2}} 1^{\frac{1/2}{1/3 + 1/2}} (15.81)^{\frac{1}{1/3 + 1/2}} \left[\left(\frac{1/3}{1/2} \right)^{\frac{1/2}{1/3 + 1/2}} + \left(\frac{1/2}{1/3} \right)^{\frac{1/3}{1/3 + 1/2}} \right]$$

$$= 1^{1/3} 1^{2/3} (15.81)^{1/3} \left[(1/2)^{2/3} + (2)^{1/3} \right]$$

$$= 299.96 \approx 300$$

Costs under initial prices, P^0 and maximal utility level, $U^1 \equiv V(P^0, w)$

$$e(P^0, U^1) = 4^{1/3} \cdot 7^{2/3} (10.90)^{1/3} \left[\left(\frac{1}{2}\right)^{2/3} + (2)^{1/3} \right] = 377.895$$

Costs under initial prices, P^1 and maximal utility level, $U^0 \equiv V(P^1, w)$

$$e(P^1, U^0) = (2)^{1/3} \cdot 4^{2/3} (15.81)^{1/3} \left[\left(\frac{1}{2}\right)^{2/3} + (2)^{1/3} \right] = 238.08$$

Costs under new prices P^1 and maximal utility level, $U^1 \equiv V(P^1, w)$

$$e(P^1, U^1) = 2^{1/3} \cdot 4^{2/3} (18.80)^{1/3} \left[\left(\frac{1}{2}\right)^{2/3} + (2)^{1/3} \right] = 299.91 \approx 300$$

iii) Equivalent variation (EV)

The amount of money that the consumer would be willing to give up before the price increase to be as well off as after the price increase.

$$EV(P^0, P^1, w) = e(P^0, U^1) - e(P^0, U^0) = 377.895 - 300 = 77.9 \approx 78$$

Compensated variation (CV):

The amount of money that a consumer would need after a price increase to be as well off as before the price increase.

$$CV(P^0, P^1, w) = e(P^1, U^1) - e(P^1, U^0) = 300 - 238.08 = 61.92 \approx 62$$

Consumer surplus (CS)

Price of commodity 1 decreased by 50%, from $P^1=4$ to $P^1=2$.

The change in consumer surplus is:

$$ES = \int_0^1 x_1(p, u) dp_1 \quad | \quad x_1(p, u) = \frac{1}{3} \frac{m}{p_1} \quad \text{where } m = 100$$

$$= \int_0^1 \left(\frac{1}{3} \frac{m}{p_1} \right) dp_1$$

$$= \int_0^1 \left(\frac{1}{3} \times \frac{100}{p_1} \right) dp_1$$

$$= 100 \int_0^1 \frac{1}{p_1} dp_1$$

$$= 69.31$$

Relationship between EV , ES and CV :

$$EV > ES > CV$$

Nature of commodities:

As $EV > ES > CV$, for the decrease in price of commodity 1, the commodities are normal goods.